

Notice that the average return is not calculated as the simple average of all returns.

Average return—logarithmic change method

The average return using the log methodology, however, is determined directly from the simple average formula. This is:

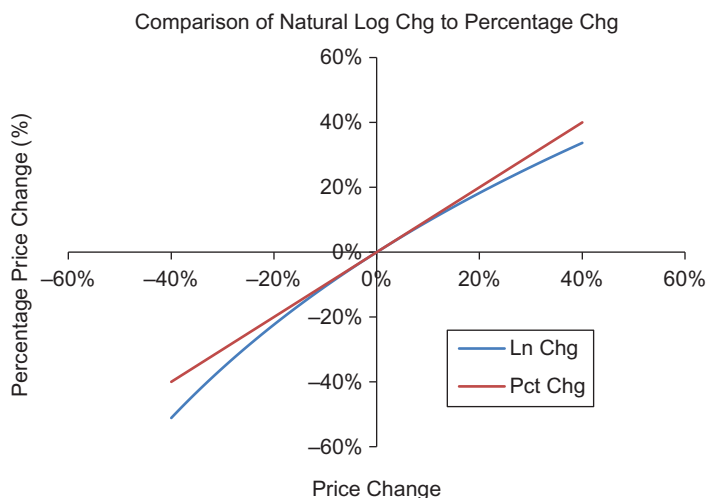
$$\bar{r} = \frac{1}{n} \sum r_k \quad (6.4)$$

The average can also be computed directly from the starting and ending price as follows:

$$\bar{r} = \frac{1}{n} \cdot \ln \left(\frac{p_t}{p_0} \right) \quad (6.5)$$

How do percentage returns and log returns differ?

For small changes, there actually is not much difference between these two measures. Figure 6.1 shows a comparison percentage price change compared to log price change over the range of returns from -40% through $+40\%$. As shown in this figure, there are very small differences in calculation process between -20 and $+20\%$, and negligible differences between -10 and $+10\%$. However, we start to see differences once we are outside of this range. For trading purposes where the daily price change is quite often less than $\pm 10\%$ either returns measure is reasonable. However, for a portfolio manager with a longer time horizon we will start to see some differences between these two measures.



■ **Figure 6.1** Comparison of Natural Log Price Change to Percentage Price Change.